

Sharp Decay Laws for the Fourier Transform of Rvachev's Up-Function

Dyadic factorization, lacunary fluctuations, extremal envelopes, and Cesàro
normalization

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Abstract

Let u be Rvachev's compactly supported C^∞ function, normalized to have integral one, and use the Fourier convention $\hat{h}(\xi) = \int_{\mathbb{R}} h(t) e^{-2\pi i \xi t} dt$. Its Fourier transform is

$$\mathcal{U}(\xi) = \prod_{n=0}^{\infty} \operatorname{sinc}\left(\frac{\pi \xi}{2^n}\right), \quad \operatorname{sinc} z = \frac{\sin z}{z}.$$

The common description “roughly log-normal” is correct only at the coarsest scale. There is no single pointwise asymptotic equivalent: $\mathcal{U}$ vanishes at every nonzero integer, and after removal of the universal quadratic logarithmic term the remainder is a lacunary Birkhoff sum for the doubling map.

This article gives an exact dyadic factorization and then separates four distinct notions of amplitude. The sharp global/extremal envelope has power exponent

$$\beta_\infty = \frac{1}{2} + \frac{\log(2\pi/\sqrt{3})}{\log 2} = 2.359014879111740 \dots$$

On almost every fixed dyadic ray $\xi = 2^k y$, the pointwise exponent is

$$\beta_{\text{typ}} = \frac{1}{2} + \frac{\log(2\pi)}{\log 2} = 3.151496129472318 \dots,$$

with sharp positive law-of-the-iterated-logarithm excursions of order $\sqrt{\log \xi \log \log \log \xi}$. The absolute Cesàro mean is controlled by the Fouvry–Mauduit constant $\lambda = 0.661322602060565 \dots$ and has exponent

$$\beta_1 = \frac{1}{2} + \frac{\log(\pi/\lambda)}{\log 2} = 2.748070014871334 \dots$$

Finally, the exponent $\log(2\pi)/\log 2$ appearing in the earlier draft is identified exactly as the L^2 (root-mean-square) exponent, not a pointwise one. A positive real-analytic, eventually decreasing gauge is constructed that rigorously satisfies the bounded-Cesàro requirement posed in the original question, and the residual dyadic profile obstructing a naive all- T limit is described by a transfer-operator eigenmeasure.

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1 Statement of the problem and the main correction

The object of study is

$$\mathcal{U}(x) = \prod_{n=0}^{\infty} \operatorname{sinc}\left(\frac{\pi x}{2^n}\right), \quad x \in \mathbb{R}. \quad (1)$$

It is real, even, entire, and rapidly decreasing on the real axis. Since $\mathcal{U}$ is C^∞ with compact support, integration by parts already gives $\mathcal{U}(x) = \mathcal{O}_N(|x|^{-N})$ for every N . What that elementary argument does not reveal is the much sharper scale of decay, nor the arithmetic oscillations hidden inside that scale.

The earlier draft proposed, in effect, a deterministic equivalent of the form

$$|\mathcal{U}(x)| \stackrel{?}{\sim} C(\log x)^{-p} x^{-\log(2\pi)/\log 2} \exp\left(-\frac{(\log x)^2}{2 \log 2}\right). \quad (2)$$

There are three immediate structural difficulties with such a statement.

1. $\mathcal{U}(m) = 0$ for every nonzero integer m , so no positive pointwise equivalent can hold on the full real axis.
2. Even on a fixed dyadic ray, the normalized remainder is a lacunary sine product. For almost every ray it has unbounded, sharply quantifiable positive logarithmic excursions.
3. Pointwise, extremal, L^1 , and L^2 amplitudes are governed by different geometric means of the same lacunary product. Conflating them produces a plausible but incorrect exponent.

The common factor in every correct notion of decay is

$$\exp\left(-\frac{(\log x)^2}{2 \log 2}\right). \quad (3)$$

The next power of x , however, depends on what “amplitude” means. The following table is a compact preview; all entries are proved below. Write

$$G_\beta(x) := x^{-\beta} \exp\left(-\frac{(\log x)^2}{2 \log 2}\right), \quad x > 0.$$

Notion of size	effective sine mean r	exponent $\beta = \frac{1}{2} + \frac{\log(\pi/r)}{\log 2}$	status of normalization
Sharp global/extremal envelope	$\sqrt{3}/2$	2.359014879...	sharp up to $x^{o(1)}$; attained on a period-two dyadic ray
Root mean square (L^2)	$1/\sqrt{2}$	$\log(2\pi)/\log 2 = 2.651496129\dots$	exact lacunary second moment
Absolute mean (L^1)	$\lambda = 0.661322602\dots$	2.748070015...	yields a bounded Cesàro ratio; dyadic-endpoint limit after scaling
Almost-everywhere pointwise	$1/2$	3.151496129...	with sharp one-sided LIL-scale lacunary excursions

The exponent in the old draft is not meaningless. It is exactly the L^2 exponent:

$$\frac{1}{2} + \frac{\log(\pi\sqrt{2})}{\log 2} = \frac{\log(2\pi)}{\log 2}.$$

What fails is its interpretation as a pointwise decay law, together with the proposed $(\log x)^{-1/2}$ correction.

2 Fourier product, zeros, and the unique extrema

2.1 Normalization and product representations

With the Fourier convention used in the abstract, the standard construction of the up-function gives (1); see Arias de Reyna [1] for a systematic treatment. The product converges normally on compact subsets of $\mathbb{C}$, since $1 - \operatorname{sinc} z = \mathcal{O}(z^2)$ near zero.

The elementary identity

$$\frac{\sin z}{z} = \prod_{m=1}^{\infty} \left(1 - \frac{z^2}{\pi^2 m^2}\right)$$

yields a second, arithmetically revealing product.

Proposition 2.1 (Integer zeros and their multiplicities). *For every $z \in \mathbb{C}$,*

$$\mathcal{U}(z) = \prod_{m=1}^{\infty} \left(1 - \frac{z^2}{m^2}\right)^{1+\nu_2(m)}, \quad (4)$$

where $\nu_2(m)$ is the exponent of 2 in m . Thus the zero at each $m \in \mathbb{Z} \setminus \{0\}$ has multiplicity $1 + \nu_2(|m|)$.

Proof. Inserting the sine product into (1) gives factors with zeros at $z = 2^n q$, $n \geq 0$, $q \geq 1$. For a fixed positive integer m , the number of pairs (n, q) satisfying $m = 2^n q$ is $1 + \nu_2(m)$. Since $\sum_{m \geq 1} (1 + \nu_2(m))/m^2 < \infty$, the collected genus-zero product converges normally and rearrangement is legitimate. $\square$

A cosine form, useful for comparison with the Fabius literature, follows from repeated use of $\operatorname{sinc}(2z) = \operatorname{sinc}(z) \cos z$:

$$\mathcal{U}(z) = \prod_{m=1}^{\infty} \left(\cos \frac{\pi z}{2^m}\right)^m. \quad (5)$$

The elementary scaling equation is

$$\mathcal{U}(x) = \operatorname{sinc}(\pi x) \mathcal{U}(x/2), \quad \mathcal{U}(2x) = \operatorname{sinc}(2\pi x) \mathcal{U}(x). \quad (6)$$

2.2 Exactly one extremum between consecutive zeros

Although the present article is mainly about amplitudes, it is useful to close a small logical gap from the original discussion: the oscillation has exactly one stationary point in every interval between consecutive positive integers.

Proposition 2.2 (Unique interlacing critical point). *For each $n \geq 1$, the restriction of $\mathcal{U}$ to $(n, n+1)$ has exactly one critical point. It is the unique maximum or minimum on that interval.*

Proof. Away from the integer zeros, logarithmic differentiation of (4) gives

$$\frac{\mathcal{U}'(x)}{\mathcal{U}(x)} = -2x \sum_{m=1}^{\infty} \frac{1 + \nu_2(m)}{m^2 - x^2}. \quad (7)$$

The series and its derivative converge locally uniformly on $\mathbb{R} \setminus \mathbb{Z}$. Put

$$A(x) := \sum_{m=1}^{\infty} \frac{1 + \nu_2(m)}{m^2 - x^2}.$$

For $x > 0$ away from the integers,

$$A'(x) = 2x \sum_{m=1}^{\infty} \frac{1 + \nu_2(m)}{(m^2 - x^2)^2} > 0.$$

On $(n, n+1)$ one has $A(x) \rightarrow -\infty$ as $x \downarrow n$ and $A(x) \rightarrow +\infty$ as $x \uparrow n+1$. Hence A has exactly one zero there, and (7) gives exactly one critical point of $\mathcal{U}$. Since the endpoints are zeros and the sign is constant in the open interval, that critical point is the unique extremum. $\square$

Let

$$E_n := \max_{n \leq x \leq n+1} |\mathcal{U}(x)|. \quad (8)$$

The sharp limsup law for E_n will follow in Section 5.

3 The exact dyadic factorization

Every $x \geq 1$ has a unique representation

$$x = 2^k y, \quad k = \lfloor \log_2 x \rfloor, \quad 1 \leq y < 2. \quad (9)$$

Write

$$t := y - 1 \in [0, 1), \quad a := \log 2,$$

and define

$$P_n(t) := \prod_{j=0}^{n-1} |\sin(\pi 2^j t)|, \quad (10)$$

$$Q_n(t) := 2^n P_n(t) = \prod_{j=0}^{n-1} |2 \sin(\pi 2^j t)|, \quad (11)$$

$$H(y) := \prod_{m=1}^{\infty} \left| \operatorname{sinc}\left(\frac{\pi y}{2^m}\right) \right|. \quad (12)$$

The tail H is continuous on $[1, 2]$, strictly positive on $[1, 2)$, and vanishes at $y = 2$ only because its first factor does.

Theorem 3.1 (Exact dyadic factorization). *For $k \geq 0$ and $1 \leq y < 2$,*

$$|\mathcal{U}(2^k y)| = 2^{-k(k+1)/2} (\pi y)^{-(k+1)} H(y) P_{k+1}(y-1). \quad (13)$$

Equivalently,

$$|\mathcal{U}(2^k y)| = 2^{-k(k+1)/2} (2\pi y)^{-(k+1)} H(y) Q_{k+1}(y-1). \quad (14)$$

Proof. Split the product (1) at $n = k$. For $0 \leq n \leq k$, set $j = k - n$. Then

$$\left| \sin\left(\pi \frac{2^k y}{2^n}\right) \right| = |\sin(\pi 2^j y)| = |\sin(\pi 2^j (y-1))|.$$

The product of the corresponding denominators is

$$\prod_{j=0}^k \pi 2^j y = (\pi y)^{k+1} 2^{k(k+1)/2}.$$

The remaining factors, indexed by $n = k + m$ with $m \geq 1$, are exactly $H(y)$. This proves (13); multiplying P_{k+1} by $2^{-(k+1)} Q_{k+1}$ gives (14). $\square$

This identity isolates the entire problem. The triangular denominator produces the quadratic logarithm. The mantissa y contributes only bounded-scale factors. All nontrivial residual behavior is carried by the lacunary product $P_n(t)$.

3.1 An exact logarithmic normal form

For a real parameter c , define

$$\Phi_c(b) := \frac{b^2}{2a} + \frac{bc}{a} - \frac{b}{2} - c. \quad (15)$$

If $L = \log x = ka + b$, then direct expansion gives the elementary identity

$$-\frac{a}{2}k(k+1) - (k+1)(c+b) = -\frac{L^2}{2a} - \left(\frac{1}{2} + \frac{c}{a}\right)L + \Phi_c(b). \quad (16)$$

Taking $c = \log(2\pi)$ in (14) gives the following exact, not merely asymptotic, decomposition.

Corollary 3.2 (Exact centered logarithm). *Let $x = 2^k y$, $1 \leq y < 2$, $t = y - 1$, and $L = \log x$. Then*

$$\log |\mathcal{U}(x)| = -\frac{L^2}{2a} - \beta_{\text{typ}}L + \Psi_{\text{typ}}(y) + S_{k+1}(t), \quad (17)$$

where

$$\beta_{\text{typ}} := \frac{1}{2} + \frac{\log(2\pi)}{a}, \quad (18)$$

$$\Psi_{\text{typ}}(y) := \log H(y) + \Phi_{\log(2\pi)}(\log y), \quad (19)$$

$$S_n(t) := \log Q_n(t) = \sum_{j=0}^{n-1} \log |2 \sin(\pi 2^j t)|. \quad (20)$$

At points where a sine factor vanishes, both sides are understood as $-\infty$.

Formula (17) is the decisive correction to a formal Euler–Maclaurin treatment of the logarithm of the sinc product. The singular function $\log |\sin|$ cannot be replaced by a smooth average with a deterministic $\log \log x$ remainder. Its dyadic samples form the Birkhoff sum $S_n(t)$, whose excursions are both arithmetically sensitive and, metrically, much larger than $\log \log x$.

4 Almost-everywhere pointwise decay and its sharp upper fluctuation

Let $\mathbb{T}(t) = 2t \pmod{1}$ be the doubling map and

$$\varphi(t) := \log |2 \sin(\pi t)|. \quad (21)$$

Then $S_n(t) = \sum_{j=0}^{n-1} \varphi(\mathbb{T}^j t)$. The classical integral

$$\int_0^1 \log |2 \sin(\pi t)| dt = 0 \quad (22)$$

shows that the Birkhoff sum is centered.

Theorem 4.1 (Typical power exponent). *For Lebesgue-almost every $t \in (0, 1)$, and hence for almost every $y = 1 + t \in (1, 2)$,*

$$S_n(t) = o(n). \quad (23)$$

Consequently, along the dyadic ray $x_k = 2^k y$,

$$\log |\mathcal{U}(x_k)| = -\frac{(\log x_k)^2}{2 \log 2} - \beta_{\text{typ}} \log x_k + o(\log x_k), \quad (24)$$

where

$$\beta_{\text{typ}} = \frac{1}{2} + \frac{\log(2\pi)}{\log 2} = 3.151496129472318798 \dots \quad (25)$$

Proof. The function φ belongs to $L^1(0, 1)$ and the doubling map is ergodic for Lebesgue measure. Birkhoff's ergodic theorem and (22) give (23). Substitute this into the exact identity (17); the bounded mantissa term $\Psi_{\text{typ}}(y)$ is absorbed into $o(\log x_k)$. $\square$

The $o(\log x)$ remainder is not remotely as small as a deterministic power of $\log x$. Aistleitner, Hofer, and Larcher proved a sharp one-sided law of the iterated logarithm for the upper excursions of precisely this lacunary product [2].

Theorem 4.2 (Sharp upper metric fluctuation). *For Lebesgue-almost every $t \in (0, 1)$,*

$$\limsup_{n \rightarrow \infty} \frac{S_n(t)}{\sqrt{n \log \log n}} = \pi. \quad (26)$$

Equivalently, for almost every fixed $y \in (1, 2)$, along $x_k = 2^k y$,

$$\limsup_{k \rightarrow \infty} \frac{\log |\mathcal{U}(x_k)| + \frac{(\log x_k)^2}{2 \log 2} + \beta_{\text{typ}} \log x_k - \Psi_{\text{typ}}(y)}{\sqrt{\frac{\log x_k}{\log 2} \log \log \log x_k}} = \pi. \quad (27)$$

Changing $\log \log \log x_k$ by the asymptotically equivalent $\log \log((\log x_k)/\log 2)$ does not alter the limit.

Proof. Theorem 2 of Aistleitner–Hofer–Larcher states that, for every $\varepsilon > 0$, $Q_{L+1}(t)$ is at most $\exp((\pi + \varepsilon)\sqrt{L \log \log L})$ for all sufficiently large L , and is at least $\exp((\pi - \varepsilon)\sqrt{L \log \log L})$ for infinitely many L , for almost every t . Taking logarithms and shifting the index gives (26). Formula (27) then follows from (17), because $k + 1 \sim (\log x_k)/\log 2$ and $\log \log(k + 1) \sim \log \log \log x_k$. $\square$

Remark 4.3. The theorem says that the multiplicative correction to the typical core is eventually at most

$$\exp\left((\pi + o(1))\sqrt{\frac{\log x}{\log 2} \log \log \log x}\right),$$

and reaches the same expression with $\pi - o(1)$ in place of $\pi + o(1)$ along a subsequence of almost every dyadic ray. These positive excursions are subpolynomial in x but larger than every fixed power of $\log x$. No symmetric assertion about the negative excursions is needed here; those are sensitive to exceptionally close returns to the integer zero set. In particular, no factor $(\log x)^c$, and especially no $(\log x)^{-1/2}$ factor, can produce a pointwise constant-amplitude normalization.

Remark 4.4 (Exceptional rays). The metric theorem does not erase arithmetic dependence. If t is dyadic rational, then an iterate of t is 0 and the ray eventually lands on integer zeros. If t is periodic under doubling, $P_n(t)$ has an exact periodic-geometric form. By constructing binary expansions with very long blocks of zeros, one can also force arbitrarily deep negative excursions in $S_n(t)$. Hence no nontrivial uniform lower envelope exists.

5 The sharp global and extremal envelope

The largest possible long-term geometric mean of $P_n(t)$ is not 1; the doubling dynamics prevents an orbit from remaining near the maximum of $|\sin \pi t|$. The exact answer is attained by the period-two orbit $1/3 \leftrightarrow 2/3$.

5.1 An ergodic-optimization lemma

Set

$$\rho := \frac{\sqrt{3}}{2}. \quad (28)$$

Lemma 5.1 (Maximum sine average). *Let $\mathcal{M}_T$ denote the T -invariant Borel probability measures on $[0, 1]$. Then*

$$\sup_{\mu \in \mathcal{M}_T} \int_0^1 \log |\sin(\pi t)| d\mu(t) = \log \rho. \quad (29)$$

The unique maximizing measure is the equidistribution on the two-cycle $\{1/3, 2/3\}$.

Proof. Put $z = \sin^2(\pi t)$. Under doubling, z evolves by the logistic map

$$R(z) = 4z(1 - z), \quad 0 \leq z \leq 1.$$

Let η be the push-forward of a T -invariant measure. If $\int \log z d\eta = -\infty$, there is nothing to prove. Otherwise invariance and $\log R(z) = \log 4 + \log z + \log(1 - z)$ imply

$$\int \log(1 - z) d\eta(z) = -\log 4. \quad (30)$$

The elementary inequality

$$z^3(1 - z) \leq \frac{27}{256}, \quad (31)$$

with equality only at $z = 3/4$, gives

$$3 \int \log z d\eta + \int \log(1 - z) d\eta \leq \log \frac{27}{256}.$$

Using (30) yields $\int \log z d\eta \leq \log(3/4)$. Since $\log |\sin \pi t| = \frac{1}{2} \log z$, this is (29). Equality forces $z = 3/4$ almost surely. Its two lifts are $t = 1/3, 2/3$, exchanged by T , so invariance forces equal masses on the two points. $\square$

Lemma 5.2 (Maximum finite products). *Let $M_n := \sup_{0 \leq t \leq 1} P_n(t)$. Then*

$$\lim_{n \rightarrow \infty} M_n^{1/n} = \rho. \quad (32)$$

Proof. The cocycle identity $P_{m+n}(t) = P_m(t)P_n(T^m t)$ implies $M_{m+n} \leq M_m M_n$, so $\lim n^{-1} \log M_n$ exists by Fekete's lemma. The standard empirical-measure argument for an upper-semicontinuous potential gives

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log M_n = \sup_{\mu \in \mathcal{M}_T} \int \log |\sin(\pi t)| d\mu(t).$$

For completeness, choose nearly maximizing t_n and pass to a weak limit of the empirical measures $n^{-1} \sum_{j < n} \delta_{T^j t_n}$. Every limit is invariant. Truncating $\log |\sin \pi t|$ from below and applying upper semicontinuity gives the upper bound. For the reverse bound, ergodic decomposition reduces matters to an ergodic invariant measure, and Birkhoff's theorem supplies a generic point. Now apply [Theorem 5.1](#). $\square$

5.2 The envelope theorem

Define

$$\beta_\infty := \frac{1}{2} + \frac{\log(\pi/\rho)}{\log 2} = \frac{1}{2} + \frac{\log(2\pi/\sqrt{3})}{\log 2} = 2.359014879111740707\dots \quad (33)$$

Theorem 5.3 (Sharp global logarithmic envelope). *One has*

$$\limsup_{x \rightarrow \infty} \frac{\log |\mathcal{U}(x)| + \frac{(\log x)^2}{2 \log 2}}{\log x} = -\beta_\infty, \quad (34)$$

where zeros contribute the value $-\infty$ and do not affect the limsup. Equivalently,

$$|\mathcal{U}(x)| \leq \exp\left(-\frac{(\log x)^2}{2 \log 2} - (\beta_\infty - o(1)) \log x\right) \quad (35)$$

uniformly as $x \rightarrow \infty$, and the exponent cannot be improved.

Proof. In (13), insert $P_{k+1}(t) \leq M_{k+1} = \rho^{k+1} e^{o(k)}$ from Theorem 5.2. Apply (16) with $c = \log(\pi/\rho)$. Since $y \in [1, 2)$ and $H(y) \leq 1$, the mantissa contribution is bounded above, giving (35).

For sharpness choose $y = 4/3$, so $t = 1/3$. The doubling orbit alternates between $1/3$ and $2/3$, and every sine factor equals ρ . Thus $P_{k+1}(1/3) = \rho^{k+1}$ exactly. Formula (13) and (16) then give

$$|\mathcal{U}(2^{k\frac{4}{3}})| = C_\star \exp\left(-\frac{(\log(2^{k\frac{4}{3}}))^2}{2 \log 2}\right) (2^{k\frac{4}{3}})^{-\beta_\infty}, \quad (36)$$

with the positive constant

$$C_\star = H(\frac{4}{3}) \exp(\Phi_{\log(\pi/\rho)}(\log \frac{4}{3})).$$

This proves the matching lower limsup. $\square$

The same exponent governs the largest local extrema.

Corollary 5.4 (Sharp envelope of the stationary values). *For E_n defined in (8),*

$$\limsup_{n \rightarrow \infty} \frac{\log E_n + \frac{(\log n)^2}{2 \log 2}}{\log n} = -\beta_\infty. \quad (37)$$

Proof. The upper bound follows from Theorem 5.3, uniformly for $x \in [n, n+1]$. For the reverse bound, put $x_k = 2^k(4/3)$ and $n_k = \lfloor x_k \rfloor$. The unique extremum in $[n_k, n_k + 1]$ has magnitude at least $|\mathcal{U}(x_k)|$. Apply (36); replacing x_k by n_k changes logarithms by $o(1)$. $\square$

Remark 5.5. This is a sharp *limsup envelope*, not an asymptotic formula for every E_n . The sporadic rises visible in numerical plots are intrinsic: the binary orbit of the mantissa can spend unusually long intervals near favorable or unfavorable parts of the sine potential.

6 Absolute mean decay and a rigorous Cesàro gauge

The original question asked for a positive analytic, eventually monotone g such that

$$\frac{1}{T} \int_{T_0}^T \frac{|\mathcal{U}(x)|}{g(x)} dx$$

remains between two positive constants, or ideally tends to one. The bounded version has a clean rigorous answer.

6.1 The Fouvry–Mauduit constant

Set

$$I_1(n) := \int_0^1 P_n(t) dt. \quad (38)$$

Fouvry and Mauduit proved [3] that there are constants $\kappa > 0$ and $0 < \lambda < 1$ such that

$$I_1(n) = \kappa \lambda^n (1 + o(1)). \quad (39)$$

Aistleitner, Hofer, and Larcher gave the rigorous enclosure

$$0.66130 < \lambda < 0.66135. \quad (40)$$

A high-precision transfer-operator computation gives

$$\lambda = 0.661322602060565 \dots \quad (41)$$

Digits beyond (40) are numerical rather than claimed here as an interval-certified proof.

The relevant transfer operator is

$$(\mathcal{L}f)(x) = \frac{1}{2} \left[\sin\left(\frac{\pi x}{2}\right) f\left(\frac{x}{2}\right) + \cos\left(\frac{\pi x}{2}\right) f\left(\frac{x+1}{2}\right) \right], \quad 0 \leq x \leq 1. \quad (42)$$

Indeed, $I_1(n) = \int_0^1 \mathcal{L}^n \mathbf{1}(x) dx$, and λ is the dominant eigenvalue of $\mathcal{L}$ on a suitable Hölder space.

6.2 The L^1 exponent and the gauge

Define

$$\beta_1 := \frac{1}{2} + \frac{\log(\pi/\lambda)}{\log 2} = 2.748070014871334 \dots \quad (43)$$

and

$$g_1(x) := x^{-\beta_1} \exp\left(-\frac{(\log x)^2}{2 \log 2}\right), \quad x > 0. \quad (44)$$

Theorem 6.1 (Bounded Cesàro normalization). *There exist constants $0 < A < B < \infty$ and $X_0 > 0$ such that, for every $T \geq 2X_0$,*

$$A \leq \frac{1}{T} \int_{X_0}^T \frac{|\mathcal{U}(x)|}{g_1(x)} dx \leq B. \quad (45)$$

Replacing T by $T - X_0$ in the denominator does not change the conclusion. Thus g_1 rigorously satisfies the weaker requirement in the original question.

Proof. Let $x = 2^k y$, $1 \leq y < 2$, $t = y - 1$, and $n = k + 1$. Apply (16) with $c = \log(\pi/\lambda)$. The choice (43) makes the powers of x cancel, and the exact factorization becomes

$$\frac{|\mathcal{U}(2^k y)|}{g_1(2^k y)} = W_1(y) \frac{P_n(t)}{\lambda^n}, \quad (46)$$

where

$$W_1(y) := H(y) \exp(\Phi_{\log(\pi/\lambda)}(\log y)) = H(y) \frac{\lambda}{\pi} y^{\beta_1-1} \exp\left(\frac{(\log y)^2}{2 \log 2}\right). \quad (47)$$

The function W_1 is bounded on $[1, 2]$ and has a positive minimum on $[1, 3/2]$.

Let

$$J_k := \int_{2^k}^{2^{k+1}} \frac{|\mathcal{U}(x)|}{g_1(x)} dx.$$

After $x = 2^k y$,

$$J_k = 2^k \lambda^{-n} \int_0^1 W_1(1+t) P_n(t) dt. \quad (48)$$

The symmetry $P_n(1-t) = P_n(t)$ gives $\int_0^{1/2} P_n = I_1(n)/2$. Hence the lower and upper bounds for W_1 , together with (39), imply

$$c 2^k \leq J_k \leq C 2^k \quad (49)$$

for all sufficiently large k , with fixed $c, C > 0$.

If $2^K \leq T < 2^{K+1}$, the sum of complete shells and the current partial shell is at most a constant times 2^K , hence at most a constant times T . For the lower bound, the complete preceding shell contributes at least $c 2^{K-1}$, whereas $T < 2^{K+1}$. This proves (45). $\square$

6.3 Analyticity and monotonicity properties of the gauge

Proposition 6.2 (Eventual alternating derivatives). *The function g_1 is positive and real analytic on $(0, \infty)$. For each fixed $m \geq 0$, there exists X_m such that*

$$(-1)^m g_1^{(m)}(x) > 0 \quad (x \geq X_m). \quad (50)$$

In particular g_1 is eventually strictly decreasing, and every fixed derivative is eventually monotone.

Proof. Put

$$u(x) := \frac{\log x}{\log 2} + \beta_1.$$

Then $g_1'(x) = -x^{-1} u(x) g_1(x)$. Induction using $\frac{d}{dx} = x^{-1} \frac{d}{d \log x}$ gives, for fixed m ,

$$\frac{g_1^{(m)}(x)}{g_1(x)} = x^{-m} \left((-1)^m u(x)^m + \mathcal{O}_m(u(x)^{m-1}) \right). \quad (51)$$

The leading term controls the sign once x is large enough. $\square$

The proposition gives eventual monotonicity separately for every derivative order. It does not assert complete monotonicity on one common half-line independent of m .

6.4 Why a simple normalization need not converge for every T

The bounded result is elementary once (39) is known. A finer spectral statement explains the remaining obstruction to a single all- T limit.

On a Hölder space, the operator (42) has a positive eigenfunction h , a positive eigenmeasure ν , and a spectral gap. Normalize $\mathcal{L}h = \lambda h$, $\mathcal{L}^* \nu = \lambda \nu$, and $\nu(h) = 1$. Then, for Hölder q ,

$$\lambda^{-n} \mathcal{L}^n q = h \nu(q) + \mathcal{O}(\theta^n \|q\|_{\text{Hol}}) \quad (0 < \theta < 1). \quad (52)$$

It follows from (48) that

$$\frac{J_k}{2^k} \longrightarrow C_1 > 0. \quad (53)$$

After multiplying g_1 by C_1 , the Cesàro average therefore tends to one along the dyadic endpoints $T = 2^K$.

For $T = 2^K y$ with fixed $1 < y < 2$, however, the newest partial shell has the same order as all preceding shells together. At continuity points of the limiting measure one obtains a profile

$$\lim_{K \rightarrow \infty} \frac{1}{2^K y} \int_1^{2^K y} \frac{|\mathcal{U}(x)|}{C_1 g_1(x)} dx = \frac{1 + D(y)/C_1}{y}, \quad (54)$$

where

$$D(y) = \left(\int_0^1 h(x) dx \right) \nu(W_1(1 + \cdot) \mathbf{1}_{[0, y-1]}), \quad D(2) = C_1. \quad (55)$$

The right side of (54) is a dyadic-mantissa profile rather than a constant.

There is a conceptual reason. The invariant equilibrium measure $\mu = h\nu$ for the potential $\log |\sin \pi t|$ is singular. Indeed, if it were absolutely continuous, the uniqueness of the absolutely continuous invariant probability for the doubling map would force it to be Lebesgue measure. But Lebesgue measure gives the normalized pressure value $-\log 2$, whereas (40) gives $\log \lambda > -\log 2$. Since h is positive and continuous, ν is singular as well. Thus the cumulative function in (55) is not linear.

The same argument applies if g_1 is multiplied by a positive continuous function of $\{\log_2 x\}$: this merely reweights the singular measure by a continuous density and cannot turn it into Lebesgue measure. Consequently, within this natural log-normal power class (even allowing smooth log-periodic corrections), one should expect a nonconstant dyadic profile rather than the full limit in the strongest version of the original question. This does not rule out deliberately scale-dependent, more exotic gauges outside that class.

7 A unified L^p spectrum

The preceding rates are instances of one pressure family. For $p > 0$, define

$$I_p(n) := \int_0^1 P_n(t)^p dt \quad (56)$$

and the weighted transfer operator

$$(\mathcal{L}_p f)(x) := \frac{1}{2} \left[\sin^p \left(\frac{\pi x}{2} \right) f \left(\frac{x}{2} \right) + \cos^p \left(\frac{\pi x}{2} \right) f \left(\frac{x+1}{2} \right) \right]. \quad (57)$$

Let λ_p be its dominant eigenvalue. Then

$$I_p(n) = \kappa_p \lambda_p^n (1 + o(1)), \quad \kappa_p > 0, \quad (58)$$

under the standard Perron–Frobenius theorem on a Hölder space. Put

$$r_p := \lambda_p^{1/p}, \quad \beta_p := \frac{1}{2} + \frac{\log(\pi/r_p)}{\log 2}. \quad (59)$$

Theorem 7.1 (L^p shell normalization). *For every fixed $p > 0$, with*

$$g_p(x) := x^{-\beta_p} \exp \left(-\frac{(\log x)^2}{2 \log 2} \right),$$

there are constants $0 < c_p < C_p < \infty$ such that, for all sufficiently large k ,

$$c_p 2^k \leq \int_{2^k}^{2^{k+1}} \left| \frac{\mathcal{U}(x)}{g_p(x)} \right|^p dx \leq C_p 2^k. \quad (60)$$

With the spectral asymptotic (58), the shell integral divided by 2^k has a positive finite limit after a constant rescaling of g_p .

Proof. Repeat the proof of Theorem 6.1, replacing λ by $r_p = \lambda_p^{1/p}$ and raising the exact ratio to the p th power. The bounded mantissa factor is positive on $[1, 3/2]$, and symmetry again supplies half of the unweighted integral on that interval. $\square$

7.1 Three exact or limiting cases

The L^1 case. Here $\lambda_1 = \lambda$, so β_1 is exactly (43).

The L^2 case. There is an elementary identity:

$$I_2(n) = 2^{-n}. \quad (61)$$

Indeed, $\mathcal{L}_2 \mathbf{1} = \frac{1}{2} \mathbf{1}$ because $\sin^2 u + \cos^2 u = 1$. Hence $\lambda_2 = 1/2$, $r_2 = 1/\sqrt{2}$, and

$$\beta_2 = \frac{1}{2} + \frac{\log(\pi\sqrt{2})}{\log 2} = \frac{\log(2\pi)}{\log 2} = 2.651496129472318798 \dots \quad (62)$$

This is precisely the power in the earlier draft.

The limits $p \downarrow 0$ and $p \rightarrow \infty$. The pressure variational principle is

$$\log \lambda_p = \sup_{\mu \in \mathcal{M}_T} \left(h_\mu(T) - \log 2 + p \int \log |\sin \pi t| d\mu(t) \right). \quad (63)$$

At $p = 0$ the equilibrium state is Lebesgue measure, and differentiation of pressure gives

$$\lim_{p \downarrow 0} r_p = \exp \left(\int_0^1 \log |\sin \pi t| dt \right) = \frac{1}{2}.$$

Thus $\lim_{p \downarrow 0} \beta_p = \beta_{\text{typ}}$. At zero temperature, (63) and Theorem 5.1 give

$$\lim_{p \rightarrow \infty} r_p = \rho = \frac{\sqrt{3}}{2}, \quad \lim_{p \rightarrow \infty} \beta_p = \beta_\infty.$$

The four headline rates are therefore not unrelated formulas; they are points on one thermodynamic spectrum.

8 Numerical evaluation of the mean constant

The rigorous interval (40) is sufficient for every qualitative and power-law conclusion above. The extra digits in (41) can be reproduced very efficiently because the two inverse branches in (42) contract by $1/2$.

Choose the Chebyshev–Lobatto nodes

$$x_j = \frac{1 - \cos(j\pi/N)}{2}, \quad 0 \leq j \leq N. \quad (64)$$

Represent a function by its values at these nodes, evaluate it at $x_j/2$ and $(x_j + 1)/2$ using barycentric interpolation, and form the $(N + 1) \times (N + 1)$ matrix corresponding to (42). Its Perron eigenvalue converges spectrally fast in this problem. A typical computation gives:

N	dominant collocation eigenvalue
8	0.6613226020798928
12	0.6613226020605659
16	0.6613226020605659
24	0.6613226020605650
32	0.6613226020605658

A compact implementation is conceptually as follows.

Listing 1: Chebyshev collocation for the Perron eigenvalue

```
x[j] = (1 - cos(pi*j/N))/2,  j = 0, ..., N
A = barycentric_interpolation_matrix(x, x/2)
B = barycentric_interpolation_matrix(x, (x+1)/2)
L = 0.5 * (diag(sin(pi*x/2))*A + diag(cos(pi*x/2))*B)
lambda_N = largest_real_eigenvalue(L)
```

For a fully certified decimal expansion, one may combine a high-precision approximate eigenfunction with a Collatz–Wielandt enclosure: if $v > 0$ pointwise, then

$$\inf_x \frac{\mathcal{L}v(x)}{v(x)} \leq \lambda \leq \sup_x \frac{\mathcal{L}v(x)}{v(x)}. \quad (65)$$

Interval arithmetic and a Chebyshev remainder bound can turn (65) into a machine-checkable enclosure.

9 Consequences and comparison with the earlier draft

The rigorous picture can be summarized in a few precise corrections.

9.1 The leading quadratic logarithm was right

Every rate contains

$$-\frac{(\log x)^2}{2 \log 2}.$$

It comes exactly from $2^{-k(k+1)/2}$ with $k \sim \log_2 x$. Thus “log-normal scale” is an accurate first description.

9.2 The next power depends on the notion of amplitude

There is no universal next coefficient. The relevant exponents are

$\beta_\infty = 2.359014879111740\dots$	sharp global/extremal envelope,
$\beta_2 = 2.651496129472318\dots$	root-mean-square amplitude,
$\beta_1 = 2.748070014871334\dots$	absolute/Cesàro mean,
$\beta_{\text{typ}} = 3.151496129472318\dots$	almost-everywhere pointwise amplitude.

The old coefficient $\log(2\pi)/\log 2$ is exactly β_2 .

9.3 The missing correction is not $\log \log x$

For typical rays, the exact deterministic algebra supplies an additional $-\frac{1}{2} \log x$ relative to the old proposed pointwise coefficient. After that, the remainder is the lacunary sum $S_n(t)$, whose sharp metric size is

$$\sqrt{\log x \log \log \log x},$$

not $\log \log x$. Consequently a fixed power of $\log x$ cannot flatten the amplitude.

9.4 Constant pointwise amplitude is impossible

There are two independent obstructions:

- exact zeros at every nonzero integer;
- unbounded positive lacunary excursions even after restriction to almost every dyadic ray.

The correct replacement is a hierarchy: sharp limsup envelope, metric pointwise law, L^p shell laws, and Cesàro bounds.

9.5 The bounded Cesàro problem is solved

The elementary analytic gauge

$$g_1(x) = x^{-2.748070014871334\dots} \exp\left(-\frac{(\log x)^2}{2 \log 2}\right)$$

satisfies the two-sided bound (45); after constant normalization, its averages converge to one at dyadic endpoints. For arbitrary endpoints, a singular dyadic-mantissa profile naturally remains.

10 Formalization-ready decomposition

The argument separates into modules of very different formal difficulty. This is useful for extending the existing Lean development.

1. **Elementary analytic module.** Formalize normal convergence of (1), the scaling law (6), the collected zero product (4), and the unique-critical-point proof Theorem 2.2.
2. **Exact dyadic algebra.** The factorization Theorem 3.1 and identity (16) are finite-product manipulations plus a tail product. This is the natural first target and yields an exact theorem with no asymptotic machinery.
3. **Ergodic optimization.** The sharp envelope reduces to subadditivity, empirical measures, and the elementary inequality $z^3(1-z) \leq 27/256$. The only substantial library component is compactness of probability measures and the variational passage for an upper-semicontinuous potential.
4. **Metric input.** The Birkhoff theorem gives the typical exponent. The sharp one-sided LIL constant π should initially be imported as an external theorem corresponding to Aistleitner–Hofer–Larcher, rather than reproved inside the Fabius development.
5. **Mean input.** Import the Fouvry–Mauduit asymptotic (39) as a named hypothesis/theorem. Once available, the proof of Theorem 6.1 is elementary shell bookkeeping. The exact L^2 identity (61) is especially attractive for immediate formalization.
6. **Transfer-operator refinement.** The dyadic endpoint limit and singular profile require Perron–Frobenius theory on a Hölder or bounded-variation space. These can be isolated from the core decay theorems and added later.

11 Conclusion

The Fourier transform of Rvachev's up-function has a universal quadratic logarithmic scale but no unique finer decay rate. The exact factorization

$$|\mathcal{U}(2^k y)| = 2^{-k(k+1)/2} (\pi y)^{-(k+1)} H(y) \prod_{j=0}^k |\sin(\pi 2^j (y-1))|$$

shows why: after the deterministic triangular denominator is removed, one is left with a lacunary multiplicative cocycle for the doubling map.

That cocycle has four natural summaries. Its maximum ergodic mean is $\sqrt{3}/2$, producing the sharp extremal exponent β_∞ . Its almost-sure geometric mean is $1/2$, producing β_{typ} together with sharp positive iterated-logarithm excursions. Its L^1 spectral radius is the Fouvry–Mauduit constant λ , producing the Cesàro exponent β_1 . Its L^2 mean is exactly $1/\sqrt{2}$, producing the exponent that appeared, without the correct interpretation, in the earlier draft.

Thus the right answer is more structured than a single corrected equivalent. The transform decays on the log-normal scale $\exp(-(\log x)^2/(2 \log 2))$, but its power correction records whether one asks for a global envelope, a typical point, an absolute mean, or a root-mean-square mean. Once those notions are kept separate, the apparent irregularity is not noise: it is the visible signature of dyadic dynamics.

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